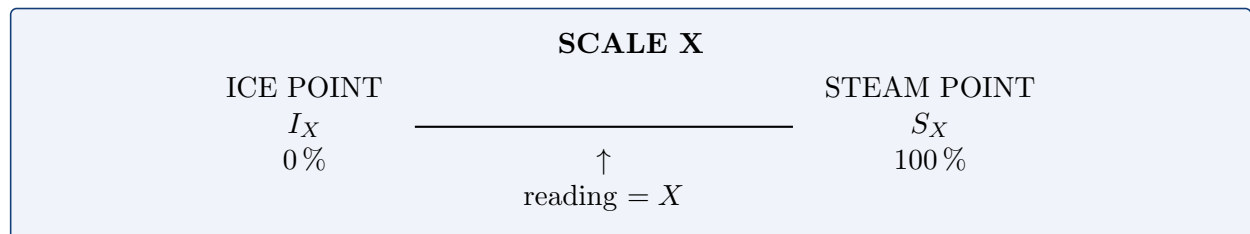


General Temperature Scale X

Class 11 Physics • Thermometry

1. The general idea

You already know Celsius and Fahrenheit. Forget them for a moment. Here is a completely new thermometer called **scale X**. It can have any numbers at the ice point and the steam point. Your only job is to *locate* a temperature on that scale.



The actual values of I_X and S_X do not matter. What matters is how far X is from the ice point, compared with the complete ice-to-steam interval:

$$\frac{X - I_X}{S_X - I_X}$$

This tells us what fraction of the ice-to-steam interval we have covered.

2. The master equation

Ice point of X = I_X	Steam point of X = S_X	Current reading = X
$\text{Fraction} = \frac{X - I_X}{S_X - I_X}$		

Comparing with Celsius:

$$\frac{X - I_X}{S_X - I_X} = \frac{T_C}{100}$$

And more generally, between X and any other linear scale Y:

$$\frac{X - I_X}{S_X - I_X} = \frac{Y - I_Y}{S_Y - I_Y}$$

3. The three numbers you must identify

Every X-scale problem gives you three important numbers:

- ① Ice point I_X
- ② Steam point S_X
- ③ Current reading X

$\text{Fraction} = \frac{\text{CURRENT} - \text{ICE}}{\text{STEAM} - \text{ICE}}$

Example. Ice point = 20°X , steam point = 80°X , current reading = 35°X .

$$\frac{35 - 20}{80 - 20} = \frac{15}{60} = \frac{1}{4}$$

The temperature is one-fourth of the way from the ice point to the steam point.

4. Type 1 — Find Celsius from X

Problem 1

A thermometer has ice point = 20°X and steam point = 80°X . What is the Celsius temperature when it reads 35°X ?

Fraction on X:

$$\frac{35 - 20}{80 - 20} = \frac{15}{60} = \frac{1}{4}$$

The same fraction must occur on Celsius:

$$\frac{T_C}{100} = \frac{1}{4} \quad \Rightarrow \quad T_C = 25^\circ\text{C}$$

Teaching point: don't look for a mysterious conversion formula for X. Just find the fraction.

5. Type 2 — Find the X reading from Celsius

Problem 2

A scale X has $I_X = 20$ and $S_X = 80$. What will it read at 40°C ?

Start with Celsius:

$$\frac{40}{100} = \frac{2}{5}$$

Therefore

$$\begin{aligned} \frac{X - 20}{80 - 20} = \frac{2}{5} & \quad \Rightarrow \quad \frac{X - 20}{60} = \frac{2}{5} \\ X - 20 = 24 & \quad \Rightarrow \quad X = 44^\circ\text{X} \end{aligned}$$

Observation. The direction has changed — earlier $X \rightarrow C$, now $C \rightarrow X$ — but the same fraction is used.

6. Type 3 — Find X from another arbitrary scale Y

Problem 3

Scale X: $I_X = 10$, $S_X = 110$. Scale Y: $I_Y = 40$, $S_Y = 240$.
Find the Y-reading corresponding to $X = 35^\circ\text{X}$.

Step 1 — fraction on X:

$$\frac{35 - 10}{110 - 10} = \frac{25}{100} = \frac{1}{4}$$

Step 2 — same fraction on Y:

$$\frac{Y - 40}{200} = \frac{1}{4} \quad \Rightarrow \quad Y - 40 = 50 \quad \Rightarrow \quad Y = 90^\circ\text{Y}$$

7. Type 4 — General algebraic scale

Problem 4

On scale X the ice point is a and the steam point is $4a$. What is the Celsius temperature when the reading is $2a$?

$$\begin{aligned} I_X &= a, & S_X &= 4a, & X &= 2a \\ \frac{2a - a}{4a - a} &= \frac{a}{3a} = \frac{1}{3} \\ \frac{T_C}{100} &= \frac{1}{3} \quad \implies \quad T_C = 33.3^\circ\text{C} \end{aligned}$$

The a has disappeared because the actual size of the scale doesn't matter. Only the fraction matters.

8. Type 5 — Negative ice and steam points

Problem 5

Scale X has $I_X = -20$ and $S_X = 80$. Find the Celsius temperature when $X = 30$.

$$\frac{30 - (-20)}{80 - (-20)} = \frac{50}{100} = \frac{1}{2} \quad \implies \quad \frac{T_C}{100} = \frac{1}{2} \quad \implies \quad T_C = 50^\circ\text{C}$$

$$-(-20) = +20$$

Whenever the ice point is negative, don't simplify it mentally — write the brackets.

9. Type 6 — Reading below the ice point

Problem 6

Scale X has $I_X = -10$ and $S_X = 90$. What Celsius temperature corresponds to $X = -60$?

$$\frac{-60 - (-10)}{90 - (-10)} = \frac{-50}{100} = -\frac{1}{2} \quad \implies \quad \frac{T_C}{100} = -\frac{1}{2} \quad \implies \quad T_C = -50^\circ\text{C}$$

A negative fraction simply means the temperature is *below the ice point*. Nothing has gone wrong.

10. Type 7 — Reading above the steam point

Problem 7

Scale X has $I_X = 10$ and $S_X = 60$. What Celsius temperature corresponds to $X = 110$?

$$\frac{110 - 10}{60 - 10} = \frac{100}{50} = 2 \quad \implies \quad \frac{T_C}{100} = 2 \quad \implies \quad T_C = 200^\circ\text{C}$$

A fraction greater than 1 is perfectly fine. It simply means we have gone beyond the steam point.

11. Type 8 — Find the unknown ice point

Problem 8

A scale X has a steam-point reading of 80°X . It reads 30°X when the actual temperature is 25°C . Find the ice-point reading.

Let the ice point be I . Since 25°C corresponds to $\frac{25}{100} = \frac{1}{4}$,

$$\frac{30 - I}{80 - I} = \frac{1}{4}$$

Cross-multiplying:

$$4(30 - I) = 80 - I \implies 120 - 4I = 80 - I \implies 40 = 3I$$

$$I = \frac{40}{3}^\circ\text{X}$$

12. Type 9 — Find the unknown steam point

Problem 9

A scale X has an ice-point reading of 10°X . It reads 40°X at 50°C . Find its steam-point reading.

Let the steam point be S . Since 50°C is halfway between 0°C and 100°C , the fraction is $\frac{1}{2}$:

$$\frac{40 - 10}{S - 10} = \frac{1}{2} \implies \frac{30}{S - 10} = \frac{1}{2} \implies 60 = S - 10$$
$$S = 70^\circ\text{X}$$

13. Type 10 — Find the unknown reading

Problem 10

Scale X: $I_X = -15$, $S_X = 65$. Find the reading on X corresponding to 40°C .

Celsius fraction: $\frac{40}{100} = \frac{2}{5}$. Therefore

$$\frac{X - (-15)}{65 - (-15)} = \frac{2}{5} \implies \frac{X + 15}{80} = \frac{2}{5} \implies X + 15 = 32$$

$$X = 17^\circ\text{X}$$

14. The general shortcut

If I_X is the ice-point reading and S_X the steam-point reading, then the X-reading corresponding to T_C is:

$$X = I_X + \frac{T_C}{100} (S_X - I_X)$$

This is just the master identity rearranged.

Example. $I_X = -15$, $S_X = 65$, $T_C = 40$:

$$X = -15 + \frac{40}{100}(65 + 15) = -15 + \frac{2}{5}(80) = -15 + 32 = 17^\circ\text{X}$$

15. A visual way to remember it

SCALE X

$$I_X \quad \text{-----} \quad S_X$$

$$\text{Total distance} = S_X - I_X \quad \text{Distance travelled} = X - I_X$$

THE GOLDEN QUESTION

“How far have I travelled from the ICE POINT?”

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“How far is the entire ICE → STEAM interval?”

$$\frac{X - I_X}{S_X - I_X}$$

16. Drill sequence

Level 1 — Direct

1. Find C from X
2. Find X from C

Level 2 — Arbitrary scale

3. $X \rightarrow Y$
4. $Y \rightarrow X$

Level 3 — Algebra

5. Ice = a , steam = $4a$, reading = $2a$
6. Ice = $2a$, steam = $5a$, reading = $3a$

Level 4 — Negative values

7. Negative ice point
8. Reading below ice point
9. Reading above steam point

Level 5 — Reverse problems

10. Find unknown ice point
11. Find unknown steam point
12. Find unknown reading

This progression teaches the structure rather than a particular numerical pattern.

17. Board summary

GENERAL SCALE X

Ice point = I_X Steam point = S_X Current reading = X

$$\frac{X - I_X}{S_X - I_X} = \text{FRACTION of the way from ICE POINT} \rightarrow \text{STEAM POINT}$$

For two linear scales X and Y:

$$\frac{X - I_X}{S_X - I_X} = \frac{Y - I_Y}{S_Y - I_Y}$$

With Celsius:

$$\frac{X - I_X}{S_X - I_X} = \frac{T_C}{100}$$

$$X = I_X + \frac{T_C}{100} (S_X - I_X)$$

Important:

- Always identify ICE, STEAM and CURRENT.
- Negative ice point $\rightarrow$ use brackets.
- Fraction $< 0 \rightarrow$ below ice point.
- Fraction $> 1 \rightarrow$ above steam point.
- In algebraic problems, arbitrary letters usually cancel.

THE KEY PHRASE

CURRENT – ICE, divided by STEAM – ICE

Once that becomes automatic, virtually every general X-scale problem becomes a one- or two-line calculation.